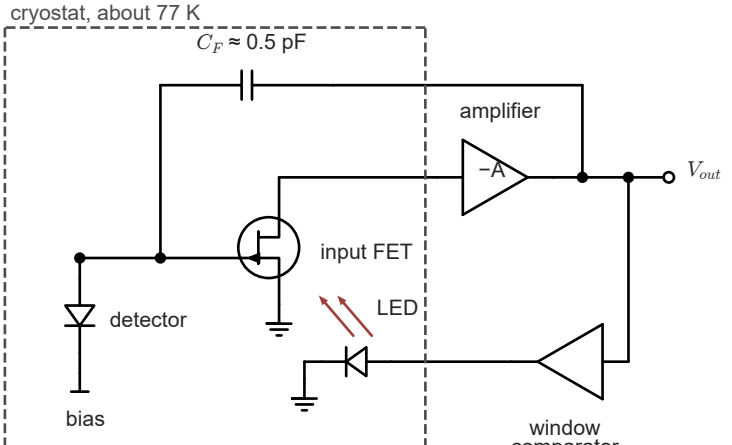
cryostat, about 77 K
$C_F \approx 0.5$ pF
amplifier
−A
V_{out}
input FET
LED
detector
bias
window
comparator